Unloop V2
Product Overview

Local-first desktop platform for real-time urge regulation

March 2026

Unloop at a glance

- Four conditions: Anxiety spike, Compulsion, Tic urge, Temptation/craving
- Three modes: Exposure (hard), Relaxed (distraction), Quickstart (emergency)
- 31+ evidence-based tasks with explainable routing and fallback logic
- Desktop-first, local-first, account-free, privacy-by-design
- Sanctuary progression system with cosmetic economy and Full Bloom rest cycles

1. Executive Summary

What Unloop is

Unloop is a local-first desktop platform for real-time urge regulation across anxiety spikes, compulsions, tic urges, and cravings. The product does not try to become a therapist, diagnosis engine, or endless chatbot. Its job is narrower and more useful: translate an active urge into a concrete next action the user can actually do now.

The Core Promise

- Translate an urge moment into a task in under a minute.
- Keep the product local-first and non-performative.
- Reward completion without turning recovery into a streak trap.
- Explain why a task was chosen and keep a fallback available.

The Four Conditions

Unloop addresses four distinct urge states, each rooted in different behavioral loops but shareable mechanisms:

Condition	Trigger Pattern	Core Mechanism	Task Approach
Anxiety Spike (ANX)	Future-focused worry, catastrophizing, or escalating physical symptoms	Avoidance + safety behavior reinforce fear	Exposure, grounding, interoceptive work
Compulsion (OCD)	Intrusive thoughts + urge to neutralize via checking, arranging, or rituals	Relief behavior strengthens the loop	ERP, semantic satiation, uncertainty tolerance
Tic Urge (TIC)	Premonitory sensation building as physical tension	Suppression backfires; completion feels relieving	Competing response, premonitory awareness
Temptation/ Craving (TMP)	Substance or behavior craving that peaks and lingers	Immediate relief reinforces impulsive action	Urge surfing, delay, stimulus control

2. The Problem

Current approaches to urge regulation work well in therapy, journaling, or reflection. But they fail at the moment when the urge is already active.

User Challenge	Current State	Unloop Advantage
Anxiety escalates faster than self-talk can keep up	Tools are generic (breathing apps, journaling) or unavailable (therapy)	Task-based exposure work that matches the anxiety type in under 1 minute
Compulsions feel relieving short-term, stronger long-term	Fragmented toolkit (notes, browser blockers, therapist homework)	Condition-specific ERP tasks that explain why resisting is mechanism
Tic urges build as physical tension, not pure thought	No condition-specific support; shame-based guidance	CBIT-informed competing response tasks tuned to the body
Cravings peak late, privately, without social support	Tools designed for retrospection, not active urge moments	Delay and urge-surfing tasks that work in the active moment

What Unloop Changes

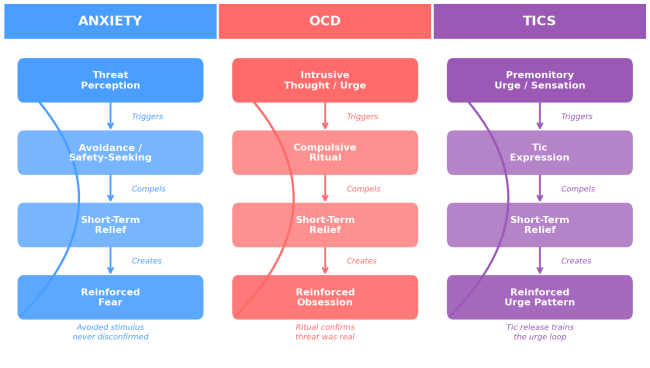
- Meets the user inside the active urge moment, not only in retrospection.
- Turns a feeling state into a task recommendation with visible reasoning.
- Keeps the product local-first and low-friction: no account, no server dependency.
- Uses progression and cosmetics to support repeat use without turning recovery into a compulsion.
- Is designed as a platform of task experiences rather than a single breathing screen.

3. Behavioral Science & Research Grounding

Unloop is grounded in peer-reviewed behavioral mechanisms, not generic wellness language. All four conditions run similar reinforcement loops, which is why the product's design principle is the same across them: help the user stay with the urge long enough for its intensity to naturally change.

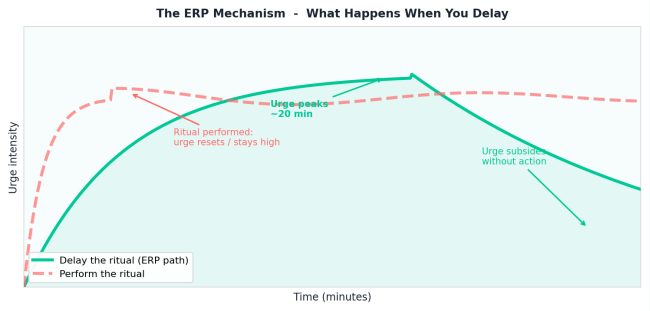
The Shared Reinforcement Loop

Anxiety, OCD, and Tics all run the same reinforcement cycle: the trigger changes, the expression changes, but the loop is structurally identical. All three use the same negative reinforcement architecture: the urge builds, the response provides short-term relief, and that relief trains the brain to run the same loop again—stronger each time. Breaking the loop requires the same move in all three cases: sit with the urge without completing the loop.



Shared Reinforcement Loop

Anxiety, compulsions, tics, and cravings follow the same cycle: trigger → urge → relief behavior → short-term relief → reinforcement.



Urge Curve Mechanism

The urge rises, peaks, and can fall without completing the ritual. Unloop's main promise: help the user stay in contact with that curve.

Evidence-Based Mechanisms by Condition

Condition	Mechanism	Why It Works
Anxiety (ANX)	Behavioral exposure, interoceptive exposure	Anxious predictions weaken when disconfirmed through direct contact with the feared state.
OCD (Compulsion)	ERP (Exposure & Response Prevention)	Relief behavior is the reinforcer. Delaying or withholding it breaks the loop and allows habituation.
Tics (TIC)	CBIT / Habit Reversal Training	Premonitory urges are physical signals. Competing responses and awareness reduce the urge without suppression.
Temptation/Craving (TMP)	Urge surfing, implementation intentions	Cravings naturally rise and fall. Observing without acting allows the curve to descend.

4. Product Architecture

Unloop is built on a clean separation: a lightweight UI layer (React + TypeScript) coordinates with a deterministic Rust backend that handles scoring, routing, and persistence.

Technology Stack

Layer	Technology	Purpose
Desktop Shell	Tauri v2	Cross-platform (Windows, macOS, Linux) with low baseline resource use.
Frontend UI	React + TypeScript + Vite	Component-based screens, fast iteration, type safety.
Backend Logic	Rust (Tauri commands)	Deterministic scoring, recommendation routing, session lifecycle, persistence.
Optional NLP	Ollama (local runtime)	On-device language understanding; disabled if runtime not present.
Persistence	Local JSON file	No cloud fallback; app works fully offline.
Styling	CSS with theme variables	Dark-first theming; tokenized color system.

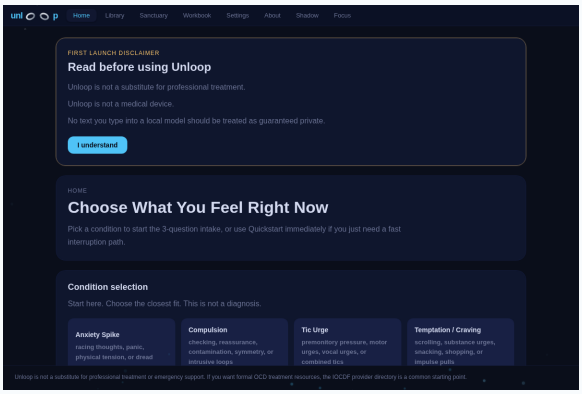
System Architecture Principles

- Screen-modular UI: Home, Task, Sanctuary, Workbook, Shadow Buddy, Focus Shield, Settings.
- Command-based backend: Frontend invokes Rust commands via secure IPC; responses are strongly typed.
- Deterministic routing: Intake scoring + mode + duration → primary task + fallback (no randomness, always explainable).
- Local-first by design: All data lives on-device; no authentication layer; full offline support.

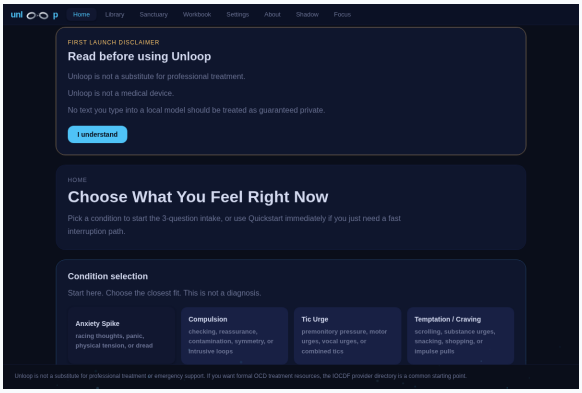
5. The Core User Experience

The user journey is intentionally short: choose what you feel → answer three quick questions → pick a mode and duration → view the task brief → complete the session.

Experience Walkthrough



Home Screen



Intake Flow

Flow Steps

1. Home: Choose Your Feeling

Four condition tiles (Anxiety, Compulsion, Tic Urge, Temptation) or free-text classifier. Quickstart always accessible.

2. Intake: Answer Three Questions

Non-specific, condition-targeted questions. Output: weighted scores for categorization and recommendation.

3. Mode & Duration Selection

Exposure (direct urge work) or Relaxed (distraction). Then 1, 3, or 5 minutes. Transparent info affordance.

4. Task Brief

'Why this task?' + instructions + fallback path. User reviews before starting the session.

5. Active Session

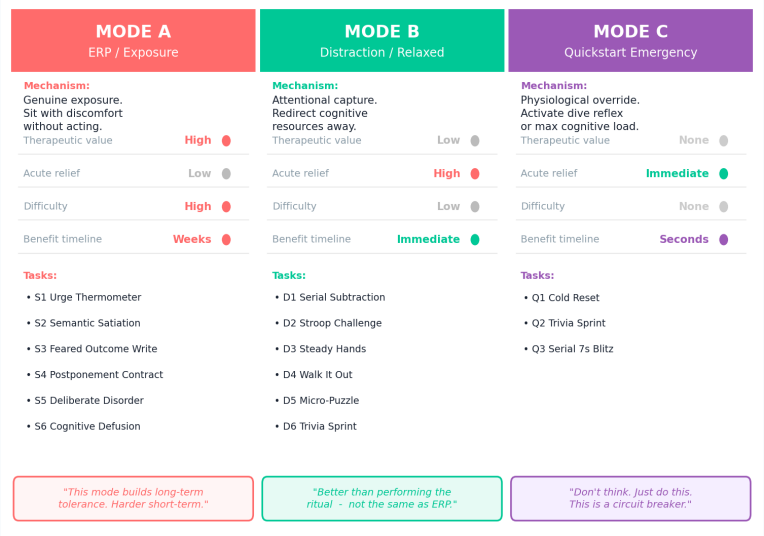
Timer runs. Task-specific mechanics (tracking, writing, observation, etc.). Fallback always available.

6. Completion & Sanctuary Reward

Session logs, Sanctuary updates, coins and XP awarded. Return to home or view Sanctuary progression.

6. Mode System

Unloop offers three distinct modes, each representing a different therapeutic tradeoff. The user chooses the mode that fits their current state.



Mode	Character	Task Selection	Use Case
Exposure (Mode A)	Direct confrontation	ERP and direct urge-work tasks	Ready to do the therapeutic work
Relaxed (Mode B)	Absorption and redirection	Attention-shift and regulation tasks	Want support without full ERP intensity
Quickstart (Mode C)	Emergency stabilization	3 condition-agnostic circuit-breakers	Crisis moment, no time for intake

Safety Constraint

App must never lock users out during an active episode. The only hard lock is Sanctuary Full Bloom 24-hour rest.

7. Task Library (31+ Tasks)

Unloop maintains a curated library of evidence-based tasks, grouped by condition and intensity. Each task includes explicit rationale and evidence labels.

Condition	Task Count	Evidence Base	Key Tasks
Anxiety (ANX)	6 tasks	Interoceptive exposure, worry postponement, grounding	Body scan, worry postponement, 5-4-3-2-1 grounding
Compulsion/OCD (S)	6 tasks	ERP, semantic satiation, uncertainty tolerance	Urge tracking, semantic satiation, deliberate disorder, uncertainty delay
Tic Urge (T)	5 tasks	Habit reversal, competing response, awareness	Competing response, premonitory tracking, tension release
Temptation/Craving (TM)	6 tasks	Urge surfing, implementation intentions, stimulus control	Urge tracking, delay windows, cost-benefit analysis
Distraction (D)	4 tasks	Attentional redirection, working-memory load	Serial 7s, Stroop task, puzzle loops, trivia sprint
Quickstart	3 tasks	Physiological reset, attention circuit-breaker	Cold reset, trivia sprint, serial 7s blitz

8. Sanctuary: Progression & Economy

Sanctuary is the primary progression anchor. It grows with every completed session, visually and mechanically, and feeds into cosmetic rewards and milestones.

Mechanic	Details
Growth Stages	7 stages unlocked at: 1, 5, 10, 18, 28, 40, 55 progress units. Each completed session advances by 1 unit.
Full Bloom Trigger	Reaches stage 7 at 55 units. Triggers 60-second celebration animation, then 24-hour mandatory rest.
Rest Cycle	During Full Bloom, home shows sleeping Sanctuary + countdown. Intake disabled; Quickstart remains accessible.
Post-Rest Reset	Progress resets to stage 1. All accessories and cosmetics persist; the growth cycle begins again.
Coins Economy	Earned per session by duration. Mode A (Exposure) earns 1.5x multiplier. Spendable only on cosmetics.
Biome Selection	User chooses one of three biomes on first launch: Tide Pool, Deep Forest, or Crystal Cave.
Accessories	Cosmetic items purchased with coins. Biome-aware; rare accessories rotate by bloom cycle.

Why This Design Prevents Compulsion

- Quickstart sessions do not advance Sanctuary (no reward for emergency use).
- Longer non-Quickstart sessions award more progress (no incentive to farm many short sessions).
- Full Bloom forces a 24-hour rest, breaking potential compulsive loops.
- Cosmetics provide intrinsic motivation without streak pressure.

9. Supporting Features

Beyond the core task engine, Unloop includes four supporting surfaces designed for specific use cases.

Feature	Purpose
Workbook	OCD-specific tools: Cycle Mapper (identify triggers and loops), Exposure Ladder (structure graded exposure work), Ritual Cost visualizer (track the true cost of avoidance). [V1]
Shadow Buddy	Anonymous paired scrapbook wall rendered with Pixi. Users can see that others are using the product without identity pressure or social comparison. Non-real-time; updates on refresh. [V1]
Focus Shield	Long-duration shielding surface with Study, Work, Sleep, and Recovery profiles. Designed for preventing triggers during high-risk periods (late-night doom scrolling, stressful work blocks, etc.). [V1]
Task Library	Direct browse-and-launch access to the full 31+ task catalog. Useful for repeat use and exploring known helpful tasks. Always updated with the latest evidence-based additions. [V2]

10. AI & Local NLP

Unloop treats optional on-device NLP as a bounded refinement tool, not a replacement for the core recommendation engine.

Design Principles

- Local-first: NLP runs on-device only; nothing sent to servers.
- Optional: If Ollama is not installed, the NLP field is hidden and the app works fully with intake scoring.
- Bounded: Model returns structured JSON with weight adjustments, never open-ended coaching or therapy chat.
- Safe: If input includes personal details, the model returns null and the user gets a gentle reminder.
- Explicit: Users see inline disclaimer and understand the local runtime requirement.

Supported Backends

Backend	Runtime	Detection
Ollama (default)	Auto-detect on 127.0.0.1:11434	App checks connection on startup; NLP field hidden if not available.
Llama Stack	Configurable endpoint (127.0.0.1:8321 default)	User manually configures in settings if preferred.

11. Privacy & Safety Design

Unloop is designed with privacy-first principles and multiple safeguards against overuse and harmful patterns.

Aspect	Approach
No Account Required	Offline-first; all data stored locally; no cloud fallback or sync. Privacy is structural, not a setting.
Minimal Text Storage	Avoid collecting sensitive personal details. Long-form writing tasks delete data after session by default.
Network Isolation	NLP backends restricted to localhost only. No outbound API calls except optional local runtimes.
Transparency	First-launch disclaimer required; never a substitute for professional treatment or medical device. Persistent info affordances.
Overuse Detection	Non-blocking warnings when >4 sessions/day, >8 app opens/day, or average spacing <45 min. Soft prompt at 5 completed sessions/day.
Full Bloom Rest	Hard lock: 24-hour mandatory rest at progression stage 7. Prevents compulsive app use; Quickstart remains accessible as anti-lockout safeguard.

12. Market Context & Positioning

Unloop fills a specific gap: real-time in-the-moment support for active urges on the desktop, grounded in clinical evidence, fully local and private.

Why Desktop-First

- Urges often strike while at the computer: browser spirals, work stress, late-night scrolling, gaming sessions.
- Desktop allows richer, longer interaction (task mechanics, device features, full writing tasks).
- Tauri keeps the app small, fast, and cross-platform without the overhead of Electron.
- Local-first design means no authentication gatekeeping—user can start immediately.

Why Local-First

- Privacy and control: no cloud vendor, no data sharing, no algorithmic feed.
- Resilience: app works entirely offline, no dependency on server uptime.
- Speed: no network latency between urge and task; recommendations instant.
- Regulatory simplicity: local persistence sidesteps medical device compliance burden.

Competitive Landscape

Category	Existing Players	Unloop Difference
Meditation/Breathing Apps	Headspace, Calm, Insight Timer	Task-routed, condition-specific, not one-size-fits-all. ERP-grounded, not just relaxation.
Mental Health Platforms	Talkspace, BetterHelp, Woebot	Offline-first, no account, no therapist middleman. For active urge moments, not ongoing therapy.
OCD/Anxiety Toolkits	NOCD (app + coaching), Beyond OCD tools	Local desktop product. Task-first, not coach-first. Works at the moment of urge, not async.
Gaming/Gamification	Habitica, Duolingo, streaks apps	Anti-compulsion design. Explicit rejection of streak traps and fear-of-loss mechanics.

13. Current Build Status

Unloop V2 is in active development on a solid proto1 foundation. The V2 shell prioritizes clarity and accessibility while preserving clinical depth.

Proto1 Validation (Historical)

- 32-task library with full device mechanics and NLP integration.
- 9-screen desktop flow validated through Playwright end-to-end tests.
- Proof-of-concept for Tauri, React, Rust command architecture, and local NLP routing.
- Strong evidence that ERP/CBIT/urge-surfing mechanics translate to interactive tasks.

V2 Build Priorities

- Calmer UI shell with tighter entry flow (home → intake → mode → task).
- Restore pixel Sanctuary renderer with three biomes and cosmetic economy.
- Rebuild task library at 31+ with explicit evidence labels.
- Preserve core mechanics: deterministic scoring, fallback routing, Full Bloom rest cycle.
- Optional local NLP with graceful degradation when runtime not present.
- Shadow Buddy, Workbook, Focus Shield as supporting screens (not core to main loop).

14. Research References & Citations

Clinical Evidence (C1-C5)

Code	Reference	Purpose in Unloop
C1	**NICE CG31: OCD and body dysmorphic disorder** https://www.nice.org.uk/guidance/cg31/chapter/1-guidance	Behavioral exposure and response prevention framework for OCD tasks.
C2	**American Academy of Neurology: Tic treatment summary** https://www.aan.com/PressRoom/Home/PressRelease/1637	Habit-reversal and CBIT grounding for tic-urge tasks.
C3	**Interoceptive exposure (PubMed)** https://pubmed.ncbi.nlm.nih.gov/39378858/	Panic and anxiety regulation via direct body-based exposure tasks.
C4	**Mindfulness-based relapse prevention meta-analysis** https://pubmed.ncbi.nlm.nih.gov/34386696/	Urge-surfing and craving-management task rationale.
C5	**Cyclic sighing and stress reduction study** https://pubmed.ncbi.nlm.nih.gov/36630947/	Physiological downshift for acute reset tasks (Cold Reset, breathing work).

Industry & Investment Thesis (P1-P4)

Code	Reference	Purpose in Unloop
P1	**YC Requests for Startups** https://www.ycombinator.com/rfs	AI workflow thesis: close decision loops, increase retention through action.
P2	**a16z AI Voice Agents 2025** https://a16z.com/ai-voice-agents-2025-update/	Voice as wedge, not whole product; AI aids urgency, not replaces human judgment.
P3	**a16z State of Consumer AI 2025** https://a16z.com/state-of-consumer-ai-2025-product-hits-misses-and-whats-next/	Retention and repeat-use framing for AI-native consumer products.
P4	**a16z Big Ideas 2026 Part 2** https://a16z.com/newsletter/big-ideas-2026-part-2/	Human-in-the-loop coordination and system-level product framing.

Technical References (T1-T6)

Code	Reference	Purpose in Unloop
T1-T5	**MDN Web APIs:** getUserMedia, enumerateDevices, Geolocation, Permissions, Web Audio	Device compatibility baseline for task mechanics (camera, mic, location, audio).

Code	Reference	Purpose in Unloop
T6	**Tauri v2 bundle configuration** https://v2.tauri.app/reference/config/#bundleconfig	Desktop packaging and cross-platform distribution reference.

Closing

Unloop is a fully-local, task-routed, evidence-grounded platform for real-time urge regulation. It exists to translate the active moment—when the urge is strongest and help is most needed—into a concrete, matched action that the user can take right now.

By grounding the product in clinical mechanisms (ERP, CBIT, urge surfing), keeping it entirely local and offline, and designing anti-compulsion safeguards into the progression system, Unloop fills a gap that generic wellness and therapy-lite apps cannot.

The desktop-first, local-first approach is not a limitation. It is the product's defining strength: it means users can turn to Unloop anywhere on their computer, without accounts, without depending on server uptime, and with confidence that their data stays private.

For more information:

Visit: unloop.co (when deployed)

Contact: [developer/investor email]

March 2026